

Summary for 2024-12-03-Duality

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1 Summary

This lecture on duality in linear programming explored the relationship between a primal linear programming problem and its corresponding dual.

*****Primal Problem:**** A primal problem is a linear programming problem formulated as a maximization or minimization problem with a given objective function and constraints.

*****Dual Problem:**** The dual problem is a related linear programming problem derived from the primal problem, where the roles of the objective function and constraints are interchanged. The dual of a maximization problem is a minimization problem, and vice versa.

*****Complementary Slackness:**** At optimality, complementary slackness conditions ensure that for each primal constraint, either the corresponding dual variable is zero or the constraint is binding (satisfied with equality). Similarly, for each primal variable, either its value is zero or the corresponding dual constraint is binding.

*****Graphic Method:**** For linear programs with two variables, the graphic method provides a geometric approach to finding the optimal solution by identifying the point within the feasible region that maximizes or minimizes the objective function.

*****Sensitivity Analysis:**** Sensitivity analysis examines how changes in the objective function coefficients (cost vector) or constraint right-hand sides (resource vector) affect the optimal solution and objective function value. Changes in the resource vector can be analyzed using shadow prices, which represent the marginal value of an additional unit of each resource.

*****Shadow Prices:**** Shadow prices, derived from the optimal dual solution, represent the marginal increase in the optimal objective function value for a one-unit increase in the corresponding resource.

The lecture demonstrated the derivation of the dual problem, the application of complementary slackness conditions, and the use of both algebraic and geometric methods to solve and analyze linear programming problems. Furthermore, it provided tools for sensitivity analysis, allowing for the assessment of the robustness of optimal solutions to changes in problem parameters. Specific examples and problem-solving techniques were used to illustrate these concepts.